

EKman Markvinnel. (2 layer)

B 1

Lösning:
$$\begin{cases} u_1 = A_1 e^{(1+i)z/n_1} + B_1 e^{-(1+i)z/n_1} + u_0 \\ u_2 = B_2 e^{-(1+i)z/n_2} + u_0 \end{cases}$$

Markvinnel:
$$\frac{\partial u_1}{\partial z} = \frac{1+i}{n_1} (A_1 - B_1) \quad (1)$$

Var:
$$A_1 - B_1 = u_0 \frac{E_1^2 - R}{E_1^2 + R} \quad \leftarrow E_1 = e^{(1+i)\tau}$$

$$R = \frac{1-E}{1+E}$$

$\Rightarrow$ Vedervinnel Θ ? $E = \frac{u_0}{v_1} \in (0,1)$, $\tau = \frac{H}{n_1}$

$\arg(xy) = \arg x + \arg y$

$$\Theta = \arg\left(\frac{1+i}{n_1} \cdot [A_1 - B_1]\right)$$

$$= \arg\left(\frac{1+i}{n_1}\right) + \arg(A_1 - B_1)$$

$$= \arg(1+i) + \arg\left(\frac{E_1^2 - R}{E_1^2 + R}\right) \quad \begin{matrix} E_1 = e^{2\tau} \cdot e^{2i\tau} \\ = e^{2\tau} (\cos 2\tau + i \sin 2\tau) \end{matrix}$$

$$= 45^\circ + \arg\left[\frac{(e^{2\tau} e^{2i\tau} - R)(e^{2\tau} e^{2i\tau} + R)}{(e^{2\tau} e^{2i\tau} + R)(e^{2\tau} e^{2i\tau} + R)}\right]$$

$$= 45^\circ + \arg\left[\frac{A^2 - R^2 + 2iAR \sin(2\tau)}{D}\right] \quad \begin{matrix} A = e^{2\tau} \\ D \in \mathbb{R} \end{matrix}$$

$$= 45^\circ + \arg\left[\frac{A^2 - R^2}{D} + i \frac{2AR \sin 2\tau}{D}\right]$$

$$= 45^\circ + \tan^{-1}\left[\frac{2AR \sin(2\tau)}{A^2 - R^2}\right] \quad (2)$$

$\arg(x+iy) = \tan^{-1}\left(\frac{y}{x}\right)$

für mit $\theta < 45^\circ$ Kreis $\tan^{-1}(x) < 0$
 $\Rightarrow x < 0$

$$(2) \Rightarrow \frac{2AR \sin(2\varphi)}{A^2 - R^2} < 0 \quad \text{ettersom } \varphi > 0 \Rightarrow A^2 > 1 \\ \text{men } R^2 \leq 1$$

$$\Rightarrow 2AR \sin(2\varphi) \quad A > 0 \quad R > 0$$

$$\Rightarrow \sin(2\varphi) < 0$$

$$\Rightarrow \varphi \in \left(\left(n + \frac{1}{2}\right)\pi, (n+1)\pi \right), \quad n=0, 1, 2, \dots$$

$$\text{Om } n=0, \quad \theta < 45^\circ \text{ om } \frac{\pi}{2} < \frac{H}{h_1} < \pi$$

Minsta möjliga vinkel? när $\sin(2\varphi) = -1$

$$\Rightarrow \varphi_{\min} = \frac{1}{2}\pi + 2n\pi \quad \leftarrow \text{för stor } \varphi \text{ ökar } A$$

$$\text{Om } n=0 \Rightarrow \varphi_{\min} = \frac{3}{2}\pi$$

lägsta vinkel om $R \rightarrow 1$ i.e. $\varepsilon \rightarrow 0$

$$\Rightarrow \theta_{\min} > 45^\circ = \tan^{-1}\left(\frac{2A}{A-1}\right)$$

$$\theta_{\min} > 45^\circ = 1,03^\circ$$

$$\theta_{\min} > 43,97^\circ$$

ÖScher

Största möjliga vinkel? kolla gränsvärde

$$\Rightarrow \varphi \rightarrow 0 \quad \text{Sätt } R=1 \text{ i.e. } \varepsilon \rightarrow 0$$

$$\theta_{\max} < 45^\circ + \tan^{-1}\left(\frac{2A}{A^2-1}\right) = 45^\circ + \tan^{-1}\left(\frac{2(1+2\varphi)^2\varphi}{1+\varphi+1-1}\right) \\ = 45^\circ + \tan^{-1}\left(\frac{4\varphi}{1+\varphi}\right) = 90^\circ$$

Slutsats:

$$\Rightarrow \theta \in (44^\circ, 90^\circ)$$

$$\Theta = 45^\circ + \tan^{-1} \left[\frac{2 R e^{2\varphi} \sin(2\varphi)}{e^{4\varphi} - R^2} \right]$$

$$\varphi = \frac{H}{n_1}, \quad \epsilon = \frac{n_2}{n_1}$$

$$R = \frac{1-\epsilon}{1+\epsilon}$$

Gränsvärde när $H \rightarrow 0$: $\Rightarrow e^{2\varphi} \rightarrow 1, \sin(2\varphi) \rightarrow 0$

$$\Rightarrow \Theta \rightarrow 45^\circ + \tan^{-1}[0] = 45^\circ$$

Gränsvärde när $\varphi \rightarrow \epsilon$ & $\epsilon \rightarrow 0$:

$$\cdot \sin(2\varphi) \approx 2\varphi \rightarrow 2\epsilon$$

$$\cdot e^{2\varphi} \approx 1 + 2\varphi \rightarrow 1 + 2\epsilon$$

$$\cdot R = \frac{1-\epsilon}{1+\epsilon} \approx 1 - 2\epsilon$$

$$\Rightarrow \Theta \rightarrow 45^\circ + \tan^{-1} \left[\frac{2(1-2\epsilon)(1+2\epsilon) \cdot 2\epsilon}{(1+2\epsilon)^2 - (1-2\epsilon)^2} \right]$$

$$= \frac{4\epsilon + 8\epsilon^2 - 8\epsilon^2 - 16\epsilon^3}{4\epsilon + 4\epsilon^2 + 4\epsilon - 4\epsilon^2}$$

$$\approx \frac{4\epsilon}{4\epsilon + 4\epsilon} = \frac{1}{1+1}$$

$$\Rightarrow \Theta \rightarrow 45^\circ + \tan^{-1}\left[\frac{1}{2}\right] = 71.565^\circ$$

$$\Theta = 45^\circ + \tan^{-1} \left[\frac{2R e^{2\varphi} \sin(2\varphi)}{e^{4\varphi} - R^2} \right]$$

$$\varphi = \frac{H}{n_1}, \quad \varepsilon = \frac{v_1}{v_2}$$

$$R = \frac{1-\varepsilon}{1+\varepsilon}$$

Grenswärde wenn $\varphi \rightarrow \sqrt{\varepsilon}$, $\varepsilon \rightarrow 0$: (effektiv $n_1 \sim \sqrt{v_1}$)

$$\bullet \sin(2\varphi) \approx 2\varphi \rightarrow 2\sqrt{\varepsilon}$$

$$\bullet e^{2\varphi} \approx 1 + 2\varphi \rightarrow 1 + 2\sqrt{\varepsilon}$$

$$\bullet R = \frac{1-\varepsilon}{1+\varepsilon} \approx 1 - 2\varepsilon$$

$$\Rightarrow \Theta \rightarrow 45^\circ + \tan^{-1} \left[\frac{2(1-2\varepsilon)(1+2\sqrt{\varepsilon})2\sqrt{\varepsilon}}{(1+2\sqrt{\varepsilon})^2 - (1-2\varepsilon)^2} \right]$$

$$= \frac{4\sqrt{\varepsilon} + 8\varepsilon - 8\varepsilon\sqrt{\varepsilon} + 16\varepsilon^2}{4\sqrt{\varepsilon} + 4\varepsilon + 4\varepsilon - 4\varepsilon^2}$$

$$= \frac{1 + 2\sqrt{\varepsilon} - 2\varepsilon + 4\varepsilon^{3/2}}{1 + 2\sqrt{\varepsilon} - \varepsilon^{3/2}}$$

$$\rightarrow \frac{1 + 2\sqrt{\varepsilon}}{1 + 2\sqrt{\varepsilon}} = 1$$

$$\Rightarrow \Theta \rightarrow 45^\circ + \tan^{-1}[1] = \underline{90^\circ}$$